

Medical Insurance Cost Regression

From exploratory data analysis to linear regression: make the model see the **BMI × smoking** interaction, then use residuals to identify what the model still misses.

Regression

1,338 rows

Linear model

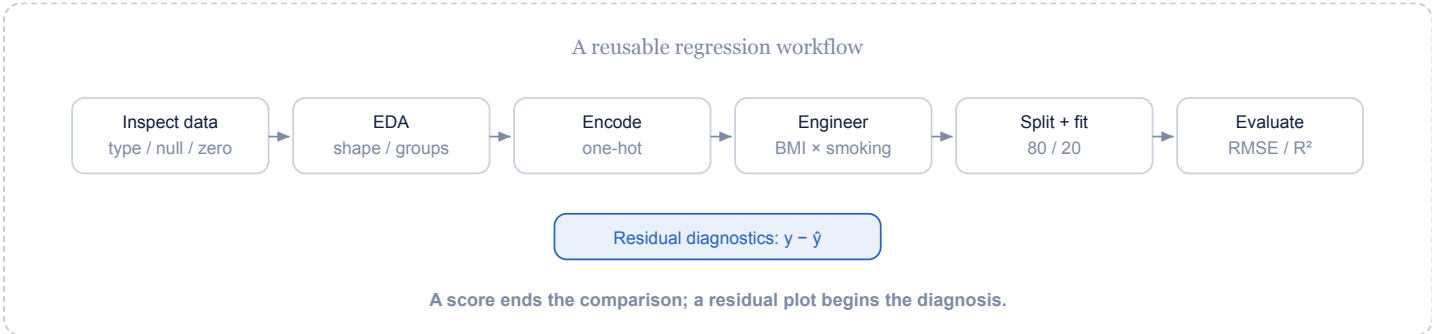
Residual diagnostics

Workflow 2.0

00 Project Task and the Complete Workflow

• First decide what kind of prediction problem you have

Target form	Task	Example
A finite set of categories	Classification	Book 08: diabetes Outcome is 0 or 1
A continuous number	Regression	This project: insurance charges



Data check: 1,338 rows, 7 columns, and no missing values. Here, **children = 0** means “no children” and is a valid observation. That differs from Book 08, where zeros in five clinical fields represented missing measurements.

Connection to Book 08: both projects follow clean → EDA → split → model → evaluate. The last branch changes: classification uses a confusion matrix, precision, and recall; regression uses MSE, RMSE, R², and residuals.

01 Define the Problem and Inspect the Data

Role	Columns	Meaning
Features, X	age , sex , bmi , children , smoker , region	Age, sex, BMI, number of children, smoking status, and region
Label, y	charges	Medical insurance cost in US dollars; a continuous value

Core concept: the model learns a mapping $x \rightarrow f \rightarrow \hat{y}$. During training it sees the true **y**. During testing it receives only **x_test**; afterward, **y_test** is compared with the held-out **y_test**. If the model sees the test answers first, the score no longer describes performance on new people.

Initial inspection

shape · types · missingness

```
insurance_data.shape
insurance_data.info()
insurance_data.isnull().sum()
(insurance_data == 0).sum()
insurance_data.describe(include="all")
```

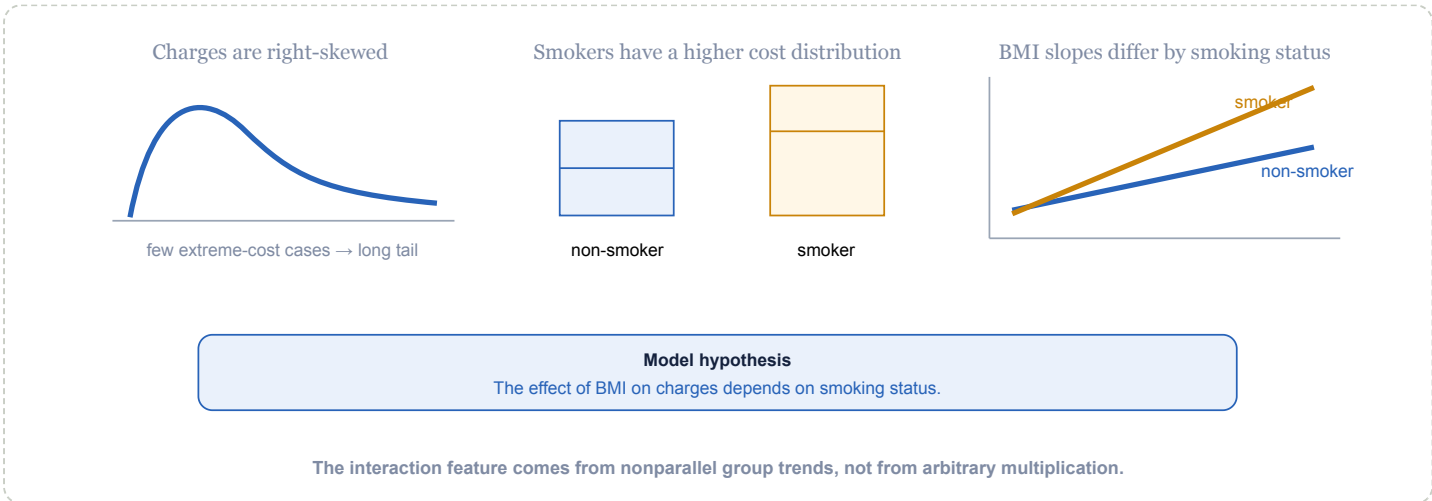
The dataset contains three categorical predictors (**sex**, **smoker**, and **region**), three numeric predictors (**age**, **bmi**, and **children**), and one numeric target (**charges**). `isnull().sum()` reports no missing values.

Field type	Can zero be genuine?	Action
Children, purchases, hospitalizations	Yes: true absence	Retain zero
BMI, age, insurance charges	Usually unreasonable	This table has no zeros; verify the source if zeros appear
Some medical measurements	Zero may encode a missing measurement	Consult the data dictionary; never impute by intuition

Why this matters: cleaning rules change the data that every later model fits. Book 08 converted some zeros to missing values because they were clinically invalid. Here, `children = 0` is meaningful. The same number can demand a different decision in a different field.

02 EDA: Derive What the Model Should Learn

Every chart should answer a model question, not merely decorate the report.



Right-skewed charges: most records have lower costs and a small number have very high costs. Squared error therefore reacts strongly to extreme cases; the meaning of a \$4,000 error changes across the cost range; and a few large residuals may reveal missing explanatory variables.

Smoking signal: a boxplot shows that smokers have a much higher median and upper distribution. It establishes a group shift, but cannot tell whether the BMI effect changes between groups. The colored scatterplot supplies that second clue.

Observed pattern	Model hypothesis	Feature
Smokers have higher costs overall	Smoking shifts the intercept	<code>smoker_yes</code>
The smoker line is steeper	The groups need different BMI slopes	<code>bmi_smoker</code>
Possible jump near BMI 30	Obesity may add a threshold effect	<code>obese</code>
Obese smokers are especially costly	Two states may reinforce each other	<code>obese_smoker</code>
The BMI pattern may curve	A smooth nonlinear term may help	<code>bmi_squared</code>

EDA → feature: feature engineering should never create columns out of thin air. Each candidate must trace back to an EDA pattern, a domain fact, or a residual-based hypothesis.

03 Encoding and Interactions: Write Hypotheses into Columns

One-hot encode categorical variables

drop_first

```
df = pd.get_dummies(  
    df,  
    columns=["sex", "smoker", "region"],  
    drop_first=True,  
    dtype=int,  
)
```

The result includes `sex_male`, `smoker_yes`, and three `region_*` columns. When all region dummies are zero, the record belongs to the omitted northeast baseline group.

Why `drop_first=True`? With an intercept, keeping both `sex_female` and `sex_male` creates the identity `sex_female + sex_male = 1`. This perfect linear dependence is the dummy-variable trap: predictions may still be computed, but coefficients are no longer uniquely and stably interpretable. In a production pipeline, fit the encoder on training data only.

One interaction term creates two BMI slopes



$$\text{charges} = \beta_0 + \beta_1 \cdot \text{bmi} + \beta_2 \cdot \text{smoker_yes} + \beta_3 \cdot (\text{bmi} \cdot \text{smoker_yes}) + \dots$$

The interaction modifies a slope; `smoker_yes` alone only shifts the whole line.

An interaction is a compact, testable statement about conditional effects.

Construct the interaction

bmi_smoker

```
df["bmi_smoker"] = df["bmi"] * df["smoker_yes"]
```

Group	smoker_yes	bmi_smoker	BMI slope
Non-smoker	0	0	β_1
Smoker	1	BMI	$\beta_1 + \beta_3$

Feature	Definition	Hypothesis	Evidence to seek
<code>bmi_smoker</code>	BMI × smoking	Different straight-line slopes	Nonparallel group scatter
<code>obese</code>	1(BMI ≥ 30)	A step change at BMI 30	A threshold or breakpoint
<code>bmi_squared</code>	BMI ²	Smooth curvature	A U-shaped residual pattern

Caution: interaction and polynomial terms should come from testable hypotheses. Blind multiplication adds complexity and overfitting risk. `obese` and `bmi_squared` are candidate modeling assumptions, not medical truths.

04 Split and Train: Make Final Evaluation Trustworthy

Fixed 80/20 split and linear regression

train · predict

```
X = insurance_data_encoded.drop(columns="charges")  
y = insurance_data_encoded["charges"]  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.20, random_state=2  
)  
  
model = LinearRegression()  
model.fit(X_train, y_train)  
y_pred = model.predict(X_test)
```

About 1,070 rows are used to learn the coefficients and about 268 rows are held out for validation. `random_state=2` does not improve the model; it keeps the sample fixed so feature versions can be compared fairly.

Step	What the model may see
<code>fit(X_train, y_train)</code>	Training features and their answers
<code>predict(X_test)</code>	Only held-out features
<code>metric(y_test, y_pred)</code>	The answer key opens only for final scoring

Three untrustworthy practices: reporting `model.score(X_train, y_train)` as final performance; accidentally keeping `charges` inside `X`; or changing the random seed for each feature version.

Q. Does ordinary linear regression require standardization?

For unregularized ordinary least squares, standardization usually does not change the fitted prediction relationship. It becomes useful for Ridge or Lasso, coefficient comparison across scales, distance-based methods, SVM, PCA, or gradient-based optimizers. If scaling is used, call `fit_transform(X_train)` and only `transform(X_test)`; never fit preprocessing on the full dataset first.

05 Regression Metrics and the Three Model Versions

Compute MSE, RMSE, and R²evaluation

```
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

Metric	Intuition	Unit	How to read it
MSE	Average squared error	US dollars ²	Penalizes large errors, but its unit is not intuitive
RMSE	$\sqrt{\text{MSE}}$	US dollars	Typical prediction-error scale
R ²	Improvement over predicting the mean	Unitless	Closer to 1 means more test-set variation is explained

If the true cost is **\$10,000** and the prediction is **\$14,000**, the residual magnitude is \$4,000 and the squared error is 16,000,000. This shows why MSE is especially sensitive to a few high-cost cases and why RMSE is easier to explain.

6,189

Baseline RMSE

4,360

Final RMSE

0.873

Final R²

Version	New information	MSE	RMSE	R ²
Baseline	Original variables + categorical encoding	—	6,189	0.745
v1	<code>bmi_smoker</code> , <code>obese</code> , <code>obese_smoker</code>	19,086,247	4,369	0.873
v2 final	v1 + <code>bmi_squared</code>	19,009,667	4,360	0.873

Main result: the largest gain comes from the v1 interaction features. BMI² adds only a marginal improvement, suggesting that most of the useful nonlinear signal has already been captured. This is a stopping signal—do not keep stacking BMI³ and BMI⁴.

Read R² correctly: `R2 = 0.873` means that on this fixed test set, the model explains about 87% of the variation in `charges`. It is not “87% prediction accuracy,” and it is not evidence that these variables cause 87% of medical costs.

Residual Plots Tell You What to Do Next

Residuals are actual minus predicted

diagnose

```
residuals = y_test - y_pred
sns.scatterplot(x=y_pred, y=residuals)
plt.axhline(0, color="red", linestyle="--")
```

A positive residual means the model predicted too low; a negative residual means it predicted too high. A healthy residual cloud is randomly scattered around zero with no visible shape.

Good: random around zero



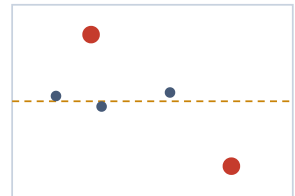
Bad: arc → missing nonlinearity



Bad: funnel → unequal variance



Bad: isolated points



Shape suggests a modeling repair; isolated points often suggest missing data or missing variables.

Signals in a bad residual plot: an arc can indicate a missing nonlinear term; a systematic group offset can indicate a missing interaction or group variable; a funnel indicates nonconstant error variance; and a few isolated points can indicate recording errors or omitted variables. “Unequal variance” corresponds to the funnel—not to a curved pattern.

Twenty-one unexplained cases: the final model still has about 21 prominent residual points. Their smoking, age, and number of children reveal no new pattern. They may require unavailable variables such as a rare disease, occupational risk, or disease severity. When residuals cluster around people whose background is absent from the data, the next step is to **collect new variables or audit samples**, not endlessly tune existing BMI features.

Reusable Modeling Template and Fair Version Comparison

Keep the project’s essential order. When reusing it, replace the plots and engineered features with hypotheses supported by the new domain’s EDA.

Complete modeling template

end to end

```
df = pd.read_csv("data/insurance.csv")
assert df.isnull().sum().sum() == 0

df = pd.get_dummies(
    df, columns=["sex", "smoker", "region"],
    drop_first=True, dtype=int
)

df["bmi_smoker"] = df["bmi"] * df["smoker_yes"]
df["obese"] = (df["bmi"] >= 30).astype(int)
df["obese_smoker"] = df["obese"] * df["smoker_yes"]
df["bmi_squared"] = df["bmi"] ** 2

X = df.drop(columns="charges")
y = df["charges"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=2
)

model = LinearRegression().fit(X_train, y_train)
y_pred = model.predict(X_test)
print(np.sqrt(mean_squared_error(y_test, y_pred)))
print(r2_score(y_test, y_pred))
```

• The correct way to compare versions

Hold y, split ratio, and random state constant

controlled comparison

```
results = []

def evaluate(name, X):
    X_tr, X_te, y_tr, y_te = train_test_split(
        X, y, test_size=0.2, random_state=2
    )
    model = LinearRegression().fit(X_tr, y_tr)
    pred = model.predict(X_te)
    results.append({
        "model": name,
        "rmse": np.sqrt(mean_squared_error(y_te, pred)),
        "r2": r2_score(y_te, pred),
    })

# Every version uses the same y, split ratio, and random_state.
```

Why compare this way? Change one feature set, record the metrics, and inspect the residuals. Only then can you say whether `bmi_smoker` helped. Mixing four changes and looking only at a better score is the regression equivalent of changing eight controls at once and then asking why precision or recall moved.

08 Common Errors and High-Frequency Questions

Common practice	Why it is wrong	Correct approach
Impute every <code>children=0</code> with the median	It changes “no children” into false information.	Read the field meaning first; retain valid zeros.
Keep both sex dummies plus an intercept	Perfect collinearity makes coefficients unstable.	Use <code>drop_first=True</code> .
Plot only BMI against charges	Smoking and nonsmoking mechanisms are mixed.	Use <code>hue="smoker"</code> .
Add only <code>smoker_yes</code>	It shifts the line but cannot change the BMI slope.	Add <code>bmi_smoker</code> when EDA supports it.
Keep adding polynomial terms after v2	A tiny test gain may be chance fit and reduce interpretability.	Check lift, residuals, and the hypothesis.
Report final R ² on <code>X_train</code>	The model has already seen those answers.	Report performance on <code>X_test / y_test</code> .
Change the seed for every split	Sampling variation contaminates version comparison.	Fix <code>random_state</code> .
Call R ² “accuracy”	Regression has no classification accuracy.	Say “proportion of variation explained relative to the mean baseline.”
Standardize all X before splitting	Test-set information enters preprocessing.	Fit transformations on training data only.

Q. Can linear regression model a curved relationship? Is it not limited to straight lines?

The model is linear in its coefficients, not necessarily in the original inputs. After adding `bmi_squared`, the prediction is still a weighted sum of coefficients, but it becomes curved along the BMI axis. The question is not “more polynomial terms are always better”; residuals or a scatterplot must support the curvature.

Q. What is the difference between `bmi_smoker` and `obese_smoker`?

`bmi_smoker` lets the BMI effect change continuously with smoking—each additional BMI unit can add a different amount for smokers. `obese_smoker` is a threshold hypothesis: smokers receive one extra fixed increment after BMI reaches 30. One changes a slope; the other changes a group-level step.

Q. Does `drop_first=True` lose information about the omitted category?

No. The omitted category becomes the reference group, represented when all related dummy variables equal zero. Coefficients change from absolute effects to differences relative to that baseline.

Q. Does RMSE = \$4,360 mean everyone is wrong by \$4,360?

No. RMSE is an overall scale. Some people may be off by only a few hundred dollars and others by tens of thousands; large misses receive extra weight. Pair RMSE with a residual plot and, when useful, group-specific errors.

Q. Is a high R^2 enough to use the model for medical or insurance decisions?

No. This table contains only a few demographic features and lacks medical history, diagnoses, treatment, and contracts; the train/test samples also come from the same source. Real pricing additionally requires external validation, fairness review, time-drift monitoring, and regulatory review.

09 Exercises, Review Order, and Checklist

• Exercises

1. **Verify what the interaction contributes.** Keep `random_state=2` fixed and run A: original + one-hot; B: A + `bmi_smoker`; C: B + `obese / obese_smoker`; D: C + `bmi_squared`. Build an RMSE/ R^2 table, explain whether B produces the largest gain over A, and use residual plots to support the claim.
2. **Compare with a nonlinear model while keeping the question clear.** Fit `RandomForestRegressor` on the same split and compare its RMSE with linear regression. If it improves, ask whether it captures the manually added interactions and thresholds or simply more complex relationships. If the gain is small, decide whether the linear model's interpretability is worth retaining.

Must be able to explain

`charges` is continuous → regression; classification accuracy, precision, and recall are not the main metrics

EDA shows the BMI–cost relationship changes with smoking → build `bmi_smoker`

`drop_first` avoids the dummy trap; all-zero dummies identify the baseline group

MSE is in squared dollars; RMSE is the typical error scale; R^2 is not accuracy or causality

A fixed split permits comparison; residual patterns suggest missing relationships, while large patternless residuals suggest missing variables

Recommended handwritten sequence

Read CSV → `get_dummies(drop_first=True)` → create interactions

`train_test_split(random_state=2)` → fit → predict

`np.sqrt(mean_squared_error(...))` + `r2_score(...)`

—

—

• Chapter checklist

- ☐ I can explain why `children=0` is not missing.
- ☐ I do not report training-set performance as the final result.
- ☐ I can move from two colored scatterplot slopes to an interaction term.
- ☐ I can explain RMSE 4,360 and R^2 0.873 without calling R^2 accuracy.
- ☐ I know that `obese` is a threshold, `bmi_squared` is curvature, and `bmi_smoker` changes a slope.
- ☐ When scores plateau, I inspect residuals instead of blindly stacking features.

One-sentence conclusion: discover relationships in the data, express them as testable hypotheses, verify improvement on a fixed test set, and accept the information limits of the dataset. The largest gain came from **BMI × smoking—the EDA-determined interaction**.